

GENERAL AGRICULTURE

Land use

Forms of land use

Forestry

- This is land set aside for growing forest trees mainly for the production of timber e.g. Chirinda forest in Chipinge

Wildlife management

- This is land allocated to wildlife so that it can grow and thrive without interfering with human occupation and human activities

Crop and livestock production

- This is the allocation of land for the production of crops and livestock to meet nation and export demands

Protected areas

- This refers to the land chosen by national authorities as scientific reserves
- Protected areas are kept under limited access to the public and are protected by law
- These areas include;
 - national parks e.g. Chimanimani, Chizarira, Gonarezhou, Hwange, Kasuma etc.
 - natural monuments e.g. Chinhoyi Caves, Victoria Falls etc.
 - nature reserves e.g. Bing Forest, Haroni/Rusitu and Vumba.
 - wildlife sanctuaries e.g. Eland, Mushundike, Kuimba and Tshabala.

Factors limiting land use

Climate

- Rainfall and temperature are the major factors influencing agricultural activities and forms of land use in Zimbabwe
- Areas receiving the same amount of rainfall are classified into natural farming regions
- Activities in each natural farming region fit the temperature and amount of rainfall received

Topography

- Terrain and slope also have an influence on agro-activities
- Mountainous areas and steep sloping areas are difficult to raise crops or rear animals
- Such areas are best used for forestry and tea production if rainfall is adequate e.g. Nyanga and Chimanimani areas

Land tenure

- Land tenure refers to the ownership, control and use of land

Forms of land tenure in Zimbabwe

Freehold tenure

- The land belongs to an individual, group or company who has property rights and full control of the land.
- The individual, group or company holds proof of ownership in the form of legal documents called title deeds

Leasehold tenure

- This is when one occupies land that is not his/hers and pays rent to the land owner for a given period of time.
- The farmer utilizing the land signs a lease contract which state the period from when the land is to be used and how the land is to be used among other things
- The owner of the land is called a *landlord* while the farmer utilizing the land is called a *tenant*

Communal tenure

- The land belongs to the state but is given to the community under leadership of the district council and traditional leaders (chiefs and headman)
- People given the land do not own the land (have no title deeds) but have user rights.

Resettlement tenure

- The land belongs to the state and is allocated to individual households according to the model of resettlement.
- Resettled farmers have to abide by rules governing the resettlement model such as keeping the required number of cattle.
- Resettlement officers and chiefs are responsible for the control of these areas and ensure that rules and regulations are followed.

Historical background to land tenure

Importance of land

- People live on land therefore it is a national heritage which can be passed on to future generations
- Land carries the most valuable mineral resources such as gold, diamond etc.
- All agricultural activities are done on land therefore land shortage negatively affects the growing of crops and rearing of animals
- Land is a habitat for wildlife which can be used as building materials, food, medicine and bring foreign currency

Pre-colonial land tenure

- Long ago people used to live a nomadic life
- They initially relied on gathering fruits and hunting of wild animals for food
- People then started domesticating the wild animals and growing crops
- They moved from place to place herding their animals as *nomads*

- Cattle, goats and sheep were the most common animals kept to provide food and clothing
- People later abandoned nomadic life in favour of a *semi-sedentary* type of life in which crops were grown under *shifting cultivation*
- Shifting cultivation is a form of agricultural practice whereby people would grow crops but did not stay permanently in one place
- Shifting cultivation involved clearing of forests and bushes, planting and caring of crops (finger and pearl millet) then moving to new land after the land had lost its fertility
- With time people changed to sedentary farming where people lived in permanent settlements and homesteads were built
- They continued to grow crops and kept animals on subsistence level up to when Zimbabwe was colonized by white settlers who then introduced commercial farming

Colonial land tenure

- The white settlers occupied Zimbabwe between 1890 to 1920
- During the period, the British South African Company (BSAC) took land cattle from indigenous people in Matebeleland and Mashonaland
- The land was occupied on racial grounds and white settlers used 45% of the total land of Zimbabwe which was in high rainfall areas and suitable for farming
- The native population were squeezed on limited land which was infertile and not fit for agricultural production

Effects of colonial rule on land ownership

- Colonial settlers secured the biggest productive land portions for themselves and their families while the majority indigenous Zimbabweans were allocated less, infertile land, which was infested with pests and diseases
- Most indigenous Zimbabweans were employed by white settlers who now owned large tracts of land
- The indigenous Zimbabweans were also forced to reduce the number of animals they owned which disempowered them
- There was very little agricultural production from the indigenous Zimbabwean community due to poor soils and retrogressive laws enacted, barring them from practicing agriculture beyond the stipulated level

Land reform: The 3rd Chimurenga

- In the year 2000, Zimbabwe embarked on the fast track land reform programme which was also called the *Third Chimurenga*
- In this programme, the disgruntled communal farmers forcibly took land from the white farmers who were owning large tracts of land at the expense of the majority of the Zimbabwean population
- The government intervened by regularizing the land redistribution process

Reasons for the land reform programme during the 3rd Chimurenga

- to correct the continued imbalances in land distribution between the blacks and the whites
- to increase agricultural productivity by peasant farmers through giving more productive land to indigenous farmers
- to create employment for the black Zimbabweans as farmers
- to involve indigenous farmers in producing export agricultural products

Resettlement models adopted during the agrarian land reform

- After land acquisition, people were settled on the pieces of land following two major resettlement models

A1 Resettlement model

- Farmers are allocated land approximately 12-15 acres for farming
- They are settled in villages headed by a village chairman
- Each village is also allocated separate grazing land in addition to the farming area
- Land in this category is governed by permits issued by the government

A2 Resettlement model

- All land under A2 model is the large-scale commercial farming
- The government issues a 99-year lease to the farm owners and the farmers are charged rentals for the land in their custody
- The farm owners also pay levies, fees and other charges as determined by the rural district council where the farm is located
- The farmer is also required to avail a 5-year development plan to the government before being allowed to lease the land

Environmental factors influencing agriculture**Temperature**

- Is the degree of hotness or coldness of a body e.g. atmospheric air or soil, it is measured in degrees Celsius (°C).
- Both atmospheric and soil temperature have an influence on crop and livestock production

Effects of temperature on plant growth**Effects of extreme high temperature on plant growth**

- Withering and dying of plants due to sun scorching
- Plants grow tall and this leads to diseases and stem lodging.
- Some plants produce seeds quickly in response to high temperature leading to poor yields.
NB: Some crops like sugarcane grow well in hot areas provided they are given sufficient water.

Effects of very low temperature on plant growth

- Slow plant growth
- Fruit injury
- Underground storage organs damage

- Flower and flower bud injury
- Tissue necrosis (death)/frost damage

Effects of temperature on water loss

- High temperature leads to high evaporation rates and evapotranspiration.
- Evaporation is a process by which liquid water changes to water vapour (gas).
- High evaporation from the ground causes:
 - i. depletion of moisture from the soil
 - ii. plant wilting, the rate of evapotranspiration may be higher than the rate of water uptake
 - iii. soil crusting due to upper layer of soil drying quickly

Evapotranspiration and wilting of crops

- Evapotranspiration is the loss of water from the soil and other water sources (dams & rivers) as well as plants through transpiration.
- Transpiration is the loss of water vapour from plants through stomata on leaves.
- Excessive evapotranspiration leads to wilting of crops.
- Wilting is the loss of turgidity in plant cells, it occurs when the rate of transpiration is higher than the rate of water absorption by plant roots.
- There are two types of wilting:

Temporary wilting

- is the loss of turgidity in plants however, the plants can recover when water is supplied.

Permanent wilting

- is the loss of turgidity in plants due to excessive water loss and the plants fail to recover even when irrigated.

Effects of temperature on livestock

Low temperature

- cause death to young animals

High temperature

- cause death of embryo in the uterus/abortion
- reduces animal appetite leading to poor animal growth
- reduce libido in male animals
- animals may come on heat without being noticed
- reduce sperm production and this leads to infertility

Humidity

- Humidity is a measure of the amount of water vapor in the atmosphere, it is measured using a hygrometer (wet and dry bulb thermometer).

Effects of high humidity

Positive effects

- slows down the rate of evaporation therefore leads to water retention in the soil
- slows down the rate of transpiration therefore prevents crop wilting.

Negative effects

- encourages growth and spread of fungal diseases in plants e.g. early and late blight in tomatoes, grey leaf spot in maize, damping off of seedlings
- encourages multiplication of pathogens in fowl runs.

Effects of low humidity

Positive effects

- less growth and spread of fungal diseases
- less multiplication of pathogens in fowl runs

Negative effects

- increased rate of evaporation and transpiration leading to wilting of plants
- reduce the rate of photosynthesis leading to slow growth rate.

Wind

- is the air in motion caused by differences in atmospheric pressure.

Importance of wind

- Required for pollination of some plants i.e. wind pollinated crops e.g. maize.
- Helps seed dispersal.
- It has a cooling effect on crop as it increases the rate of transpiration.

Effects of wind

- Strong winds damage flowers, break tree branches and cause lodging of crops therefore reduce yield.
- It increases the rate of evapotranspiration and therefore crop wilting.
- Spread pests and diseases.
- Causes soil erosion.

Rainfall

- is precipitation in the form of water droplets released by clouds.

Effects of excessive rainfall

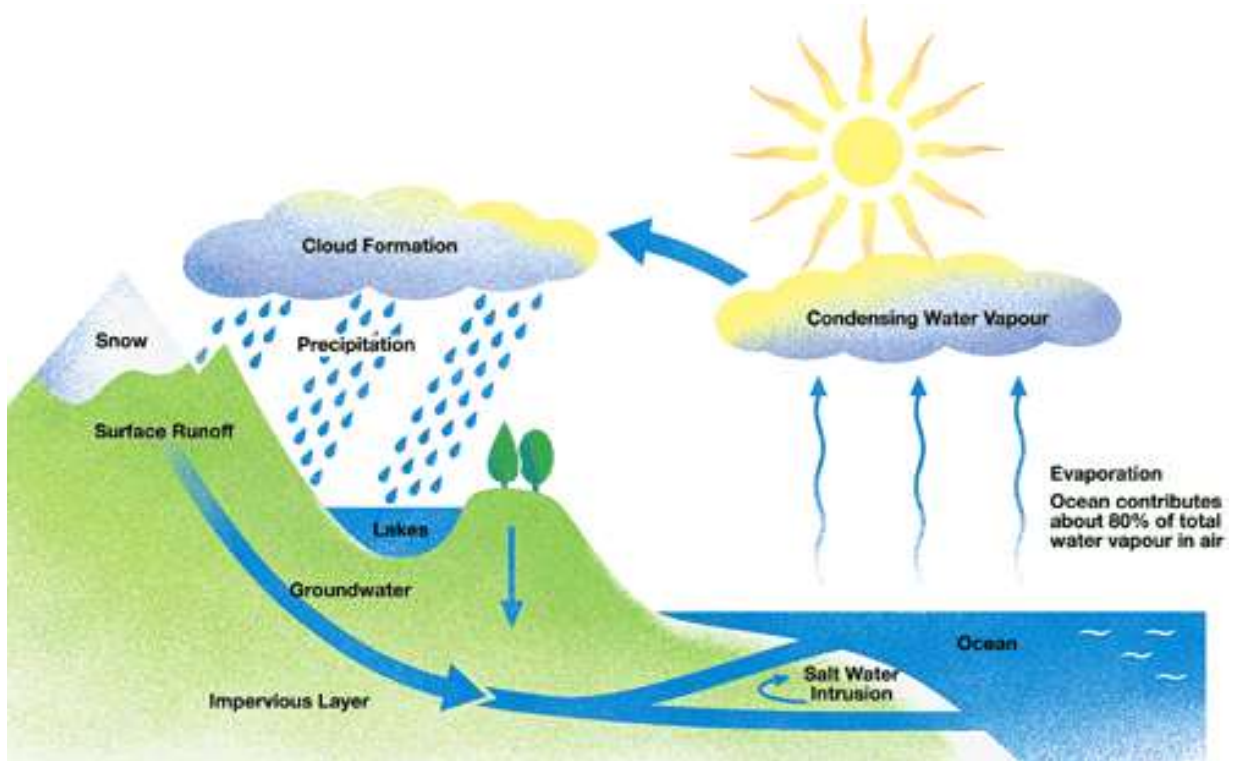
- soil erosion
- plant lodging
- leaching of nutrients
- water logging leading to root damage and multiplication of diseases.

Effects of very low rainfall

- lack of enough soil moisture to support plant growth
- permanent wilting and death of plants
- little grain growth leading to reduced yield
- total crop failure resulting in starvation.

The water (hydrological) cycle

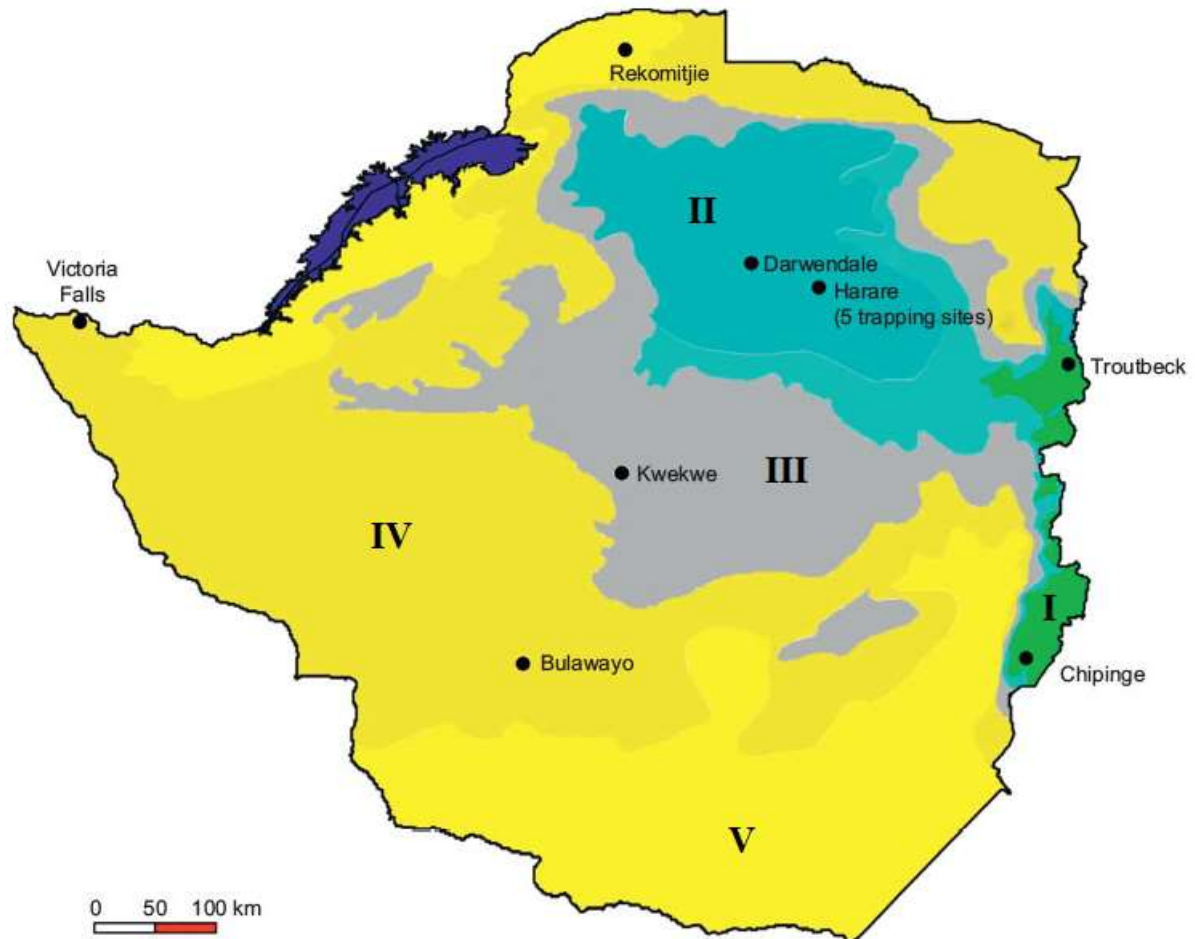
This is the cyclical movement of water through transformation from the earth into the atmosphere and back.



- Water evaporates from water bodies, trees transpire and animals perspire, this release water vapour into the atmosphere.
- The water vapour rises and condenses to form clouds which later result in precipitation reviving water sources on the earth's surface.
- Water from the soil is used by plants and animals which later transpire, perspire and the water evaporates repeating the process over and over again.

Natural farming regions

- Rainfall, temperature and soil quality were used to divide the country into the natural farming regions



Natural farming region	Annual rainfall	Annual temperature	Agricultural activities and farming systems
I	Above 1000mm	Mean annual - 15-18°C Mean minimum - 10-12°C Mean maximum - 19-23°C	• Suitable for dairy, forestry, tea, coffee, beef and fruit production • Horticultural crops such as potatoes, vegetables and protea flower production
II	750-1000mm reliable rains, falling between November to March/April	Mean annual - 16-19°C Mean minimum - 10-13°C Mean maximum - 19-23°C	• Suitable for intensive crop and animal production • Crop production includes tobacco, maize and cotton production as well as irrigated wheat and barley in winter • Livestock production on pastures and pen fattening includes beef, pig and poultry
III	500-700mm with mid-season dry spells	Mean annual - 18-22°C Mean minimum - 11-15°C Mean maximum - 23-26°C	• Suitable for drought tolerant crops and semi-intensive livestock production on fodder feed • Common crops grown are maize, cotton, groundnuts and sunflower
IV	450-650mm with severe dry spells and frequent seasonal droughts	Mean annual - 18-24°C Mean minimum - 11-20°C Mean maximum - 23-26°C	• Suitable for the production of drought resistant crops such as sorghum, pearl millet, finger millet and maize • Cattle production under semi-extensive system and wildlife production
V	Below 560mm and very erratic	Mean annual - 21-25°C Mean minimum - 14-18°C Mean maximum - 26-32°C	• Unsuitable for dry land crop production • Suitable for extensive cattle and goat production • Cattle and game ranching

Forestry

Forestry is the practice of planting, caring and managing use of trees.

Importance of forests

Economic importance

- raw materials for paper, cloth, rubber industries and used for making other wearing materials (Baobab, Mupfuti, Munhondo and Msasa)
- construction poles and building materials e.g. trusses, doors and frames
- chemicals e.g. dyes and tannin
- timber from wood is used to make furniture e.g. kitchen units, tables, chairs and wardrobes.
- creates employment
- source of foreign currency
- trees provide plywood railroad sleepers

Cultural importance

- food in the form of fruits or roots
- shelter for people and animals
- firewood
- medicinal herbs (roots, barks and leaves)

Ecological importance

- control surface runoff i.e. reduces soil erosion
- recreation, tourism, national parks, protection of endangered species (provides natural habitat for wild animals)
- wind breaks
- trees transpire and release water vapour to bring rainfall
- trees use carbon dioxide in the atmosphere

Major forests in Zimbabwe

1. Binga forest
2. Mutare North indigenous forest
3. Wattle forest in the Eastern Highlands
4. Mvuma Eucalyptus forest reserve

Indigenous and exotic trees grown in Zimbabwe

Indigenous timber trees

Name	Properties and uses
Blood wood (Mukwa)	○ A durable hard wood ○ Used for making furniture and floor tiles
Teak	○ A durable hard wood ○ Used for making floor tiles, railway planks and furniture

Mahogany (Mukusi)	○ A durable hardwood ○ Used for making furniture, floor tiles and railway line planks
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Exotic timber trees

Name	Properties and uses
Pine trees	○ High strength ○ Easy to cut and nail
Gum trees (Eucalyptus)	○ Should be treated to last long ○ Used for making electricity and telephone poles as well as firewood

Wildlife

Wildlife refers to all forms of plants (flora) and animals (fauna) in their natural environment

Flora and Fauna found in Zimbabwe

Flora (indigenous plants)	Fauna (undomesticated animals)
• Mukwa • Mahogany • Teak • Flame lily • Aloe • Musasa • Munhondo	• Monkeys • Zebra • Pangolin • Cheetah • Python • Crocodile • Eagle

Importance of wildlife

Socio-economic importance

- Earns foreign currency through tourism and sales of wildlife products such as hides and tusks (ivory)
- Creates employment in game parks, hotels and travel agencies
- Source of revenue for hotels and airlines
- Important for safari, hunting and entertainment

Cultural importance

- Wild animals are used as totems (people of the same totem are related and normally do not marry each other)
- Sight of certain animals is believed to be a sign of good or bad omen
- Skins of some wild animals are worn by traditional leaders to show their respect their ancestors and the society at large.
- Change of behaviour patterns of certain wildlife are associated with changing seasons
- Wildlife improves the aesthetic value of a place

Ecological importance

- Wild animals are part of the ecosystem, their waste improves soil fertility therefore benefits plants

Classification of wildlife according to feeding habits

Producers (autotrophs)

- This group consists of organisms capable of manufacturing their own food e.g. green plants
- The energy stored by plants is transferred when animals feed on them and the energy is then converted into animal products

Herbivores

- Herbivores are animals that feed on plant material (producers)
- They are further divided into:

Grazers

- Animals that mainly feed on grass e.g. wild beast, zebra, buffalo etc.

Browsers

- Animals that mainly feed on trees (flowers, fruits, twigs and leaves) e.g. impala, giraffe, kudu, elephant etc.

Carnivores

- These animals feed on meat (flesh) only
- They hunt and kill their prey for food e.g. lion, leopard, cheetah, hyena, eagle etc.

Omnivores

- These animals feed on both plant and animal material e.g. baboons and monkeys

The “big five”

- In Africa five animals were selected into the “big five” category
- The grouping was named by hunters who identified them as the most difficult to hunt
- The big five animals are:
 1. Lion
 2. Elephant
 3. Buffalo
 4. Leopard
 5. Rhinoceros
- These animals are found in national parks and game reserves in Zimbabwe

SOIL AND WATER

Soil formation

- Soil is the loose layer of material covering the earth's surface
- It is the medium in which plants grow.
- The soil was formed through the weathering of rocks.
- The rock from which soil is formed is called *parent rock*.

Types of rocks from which the soil is formed

Soil is formed from the following types of rocks:

Igneous rocks

- These are rocks formed from the cooling and solidification of molten lava/magma e.g. granite, basalt etc.

Sedimentary rocks

- Rocks formed from materials produced during weathering of igneous rocks e.g. sandstone, limestone, ironstone, coal etc.

Metamorphic rocks

- These are formed either from igneous or sedimentary rocks by the action of heat, pressure and chemical changes e.g. quartzite, slate, marble etc.

The role of weathering in soil formation

- Weathering is the process whereby rocks are broken down into smaller particles (soil).
- Agents of weathering are water, wind, temperature, carbonic acid, oxygen, hydrogen, plants and animals.
- Weathering is mainly influenced by climate and type of parent rock
- Different types of parent rocks weather differently and yield different types of soils.
- There are three types of weathering and these are:
 - Physical weathering
 - Chemical weathering
 - Biological weathering

Soil texture

- Soil texture is the percentage composition of sand, silt and clay particles in a soil.
- Soil particles found in the soil are called *soil fractions*
- The soil fractions are classified according to size measured by their diameter

Soil fractions

Soil fraction	Particle size (diameter)
Stone	>20mm
Gravel	2.0mm-200mm
Sand	0.02mm-2.0mm
Silt	0.002mm-0.02mm
Clay	<0.002mm

The significance of soil texture to plant growth

- It influences the type of crops to grow e.g. groundnuts must be grown in light textured soil for easy pulling at harvesting.
- It influences water holding capacity of the soil, heavy textured soils have higher water holding capacity than light textured soils.
- It determines fertilizer application rate and frequency e.g. sandy soil is highly leached therefore split application of high quantity fertilizer is recommended.
- It influences the type of irrigation method e.g. sandy soils are not suitable for surface irrigation methods because they are highly drained and the water cannot be distributed evenly

Soil structure

- Soil structure refers to the arrangement of soil particles (sand, silt and clay) into clusters called aggregates.

Types of soil structure

Crumb structure

- The aggregates are granular and porous
- It has a mixture of sand and clay
- It contains organic matter and has a good water-holding capacity
- It is also high in nutrients and well aeration
- It allows easy root penetration
- This is the best soil structure for crop production

Single grain structure

- The particles are not aggregated
- It is made up mainly of sand particles
- The soil has poor water holding capacity and poor nutrient status
- It requires heavy application of fertilizer

Soil fertility

Plant nutrients

- These are elements needed by plants for growth and reproduction
- There are two main groups of plant nutrients: major and minor elements.

Major plant nutrients (macro-nutrients)

Nutrient	Functions in plants	Deficiency symptoms	Over supply or toxicity
Nitrogen (N)	• Stimulates plants to grow vigorously • It is part of chlorophyll involved in photosynthesis • It is also required for protein formation	• Stunted growth • Yellowing of leaves (chlorosis) and plants fail to photosynthesise due to lack of chlorophyll	• Delayed maturity and fruit ripening • Plant lodging due to rank growth • Reduce yield quality e.g. high protein content in barley • Scorching of leaves
Phosphorus (P)	• Stimulates root development • Stimulates flower formation • Uniform crop maturation • Normal seed development • Photosynthesis, respiration and cell division	• Stunted root development • Purpling of leaves • Poor flower formation and delayed fruit ripening • Poor nodulation in legumes • Small and few tuber production	• Early crop maturity • Early and accelerated fruit ripening
Potassium (K)	• Increase plant resistance to lodging and diseases by strengthening stalks and stems • Increase resistance of plant to frost • Improves quality of fruits • Activation of enzymes • Protein synthesis • Improves drought tolerance in plants	• Marginal necrosis (burnt leaf edges) • Curling of leaves • Leaves fall off prematurely	• Causes magnesium deficiency • Causes luxury consumption resulting in reduced yield and quality
Calcium (Ca)	• Cell wall formation and cell division • Seed and fruit formation • Pod formation.	• Results in the formation of empty pods (<i>pops</i>) in groundnuts • Rolling up of leaves • Chlorosis along leaf margins • Poor growth of terminal buds	
Magnesium (Mg)	• Chlorophyll formation and photosynthesis • Normal cell division • Activation of enzymes and respiration	• Yellowing of older leaves • Premature leaf fall	
Sulphur (S)	• Protein synthesis • Vitamin and enzyme formation • Promotes nodulation in legumes	• Uniform chlorosis along leaf veins • Poor nodulation in legumes	

Minor plant nutrients

Nutrient	Functions in plants	Deficiency symptoms	Over supply or toxicity
Boron (B)	- Cell division and growth of meristematic tissues, flowers and vascular bundles - Pollen grain germination and pollen tube growth (pollination) - Normal fruit setting and fruit formation	- Distortion of leaves and necrosis of stems - Irregular cob filling in maize - Development of heads with many hollow hulls in sunflower - Formation of hard fruits	Leaf tips turn brownish followed by rapid necrosis of the whole leaf
Copper (Cu)	- Growth of plant tips - Catalyses plant processes (respiration, photosynthesis and protein synthesis)	- Death of plant tips - Retarded plant growth	
Manganese (Mn)	- Required for chlorophyll formation - Protein synthesis - Important in respiration	Intervinal chlorosis mainly in older leaves	
Molybdenum (Mo)	- Essential for nitrogen fixing bacteria in root nodules of legumes - Seed germination and seedling development	- Uniform paling and floppy appearance of young plants - Whip tail in brassicas such as broccoli and cauliflower	
Zinc (Zn)	Enzyme and chlorophyll formation	- Reduced growth - Chlorosis - Leaf stripping	
Cobalt (Co)	Nitrogen fixation in legumes	Poor growth in legumes	
Chlorine (Cl)	Promotes water uptake and oxygen production during photosynthesis	Poor water uptake and reduced photosynthetic rate.	

Soil and water conservation

Soil erosion and conservation

- Soil erosion refers to the removal of top soil from the earth's surface due to the agents of soil erosion
- Agents of soil erosion are run-off (running) water, raindrops and wind
- These agents are more active when the soil is bare and not covered with organic matter
- Agents of soil erosion influences the type of erosion

Types of soil erosion

Sheet erosion

- This is the *uniform* removal of a top thin layer of soil by surface run-off
- Sheet erosion encourages other types of erosion to occur

Rill erosion

- It is when very small and defined channels (*rills*) of flowing water form on the soil surface
- Rills frequently occur as straight lines between rows of crops along tillage marks usually on recently cultivated land which is not covered
- If not controlled, rills develop into gullies

Gulley erosion

- Gulley erosion occurs when rills are left unchecked and gather more water cutting the soil into the channel which gradually becomes deeper, wider and longer resulting in *gully formation*
- Gully erosion cuts deep into the sub soil and reduces arable land as the gullies continue to grow wider and deeper

Causes of soil erosion

Deforestation

- Deforestation results in the removal of vegetative cover on the soil surface
- If the soil surface is left with little or no vegetative cover, the soil becomes prone to erosion as it is exposed to agents of soil erosion
- Plant roots improve soil aggregation which reduces run-off and improve infiltration
- Bare soils as a result of deforestation become heavily compacted and reduce infiltration causing more run-off which increases soil erosion

Ploughing up and down slopes

- Surface run-off is increased if a farmer ploughs along the slope causing it to follow the slope thereby deepening and widening the channels carrying away soil which has been loosened

Stream bank cultivation

- Ploughing within 30 metres of a stream or river cause loosened soil to be easily carried away by running water down the slopes in the streams or rivers during rain storms

Monoculture

- This is when a farmer grows one crop in the same piece of land, year after year
- This results in the soil being overused and loose causing it to become easily eroded

Overstocking

- This is when a farmer keeps more livestock than the carrying capacity of the pasture land
- This results in overgrazing where grass whose roots hold soil particles together is destroyed
- In addition, hooves of animals such as cattle, goats and donkeys loosen the soil thus reduces its resistance to erosion

Poor citing of farm road

- Poorly cited farm roads especially on depressions become water channels and increase the removal of soil by flowing water

Removal of crop residues

- Removing crop residues after harvesting leaves the soil unprotected and prone to erosion

Veld fires

- Uncontrolled burning of the veld destroys vegetation and humus that protects the soil

Consequences (effects) of soil erosion

Soil loss

- Soil erosion removes top soil cover leaving most of the plant roots exposed to sun scotch

Reduced crop yields

- When top soil is washed away from the soil surface, nutrients which support crop growth are also removed
- Soil erosion therefore results in infertile soil with inadequate nutrients for good yields

Water pollution

- Surface run-off carrying mud and sediments from eroded areas ends up in rivers and dams
- Muddy water is not ideal for domestic and industrial use
- Fertilizers and agro-chemicals may be dissolved in running water and is carried with soil into streams, rivers and dams
- The nutrients encourage the development of weeds in dams and rivers e.g. water hyacinth (eutrophication)
- Agro-chemicals that find their way into rivers and dams results in death of aquatic life

Land degradation

- Unchecked soil erosion results in the development of gullies in fields thus reduces the size of arable land as the soil becomes unsuitable for vegetation growth

Siltation of rivers and dams

- The soil washed away by erosion is deposited in rivers and dams
- Deposition of soil reduces the volume of water these water sources can hold resulting in less water available for agricultural, domestic and industrial uses

Prevention and control of soil erosion

Biological methods

In arable lands

- Early planting for early plant establishment
- Strip cropping
- Intercropping/mixed cropping

In grazing lands

- Correct carrying capacity in paddocks
- Rotational grazing
- Avoid uncontrolled burning of grass
- Fencing of eroded areas so as to protect them from being grazed

Mechanical methods

- Terracing
- Construction of tied ridges
- Construction of storm drains and contour ridges
- Construction of wind breaks
- Contour ploughing
- Minimum tillage (zero tillage)

Cultural methods

- Practice rotational grazing
- Avoid veld fires
- Addition of organic matter
- Correct fertilizer application
- Mulching
- Addition of agricultural lime so that soils will form aggregates leading to a good crumb structure
- Crop rotation
- Maintaining optimum plant populations

Water conservation

- This is the wise management and utilization of water as a natural resource in a manner that does not cause shortage in the future.

Importance of water conservation

- Water conservation increases crop yields through the availability of moisture during critical growth stages of crops
- It also helps the survival of soil organisms that decompose and contribute to nutrient recycling
- Water conservation ensures the availability of water for agricultural, domestic and industrial purposes

Methods of water conservation in arable lands

Mulching

- Mulch cover the soil surface and reduces the rate of evaporation
- Crop residues such as maize, sorghum and millet stalks can be left on the soil surface to act as mulch
- The presence of mulch also reduces runoff and improves the rate on infiltration and water storage in the soil

Contour ploughing

- Contours are dug across steep slopes to minimize the loss of surface water which may occur when water runs down the slope after heavy rains
- The contours promote infiltration of water into the soil

Terracing

- Terraces help to control surface runoff by holding water on the surface and promoting infiltration

Strip cropping

- Crops are grown in strips across the general slope of the land
- Each strip is at right angle to the flow of water and this blocks water flow to adjacent strips or rows of crops

Deep ploughing

- Deep ploughing reduces surface runoff thus encourages infiltration

Minimum tillage

- When soil is not disturbed, the rate of evaporation is reduced and more water will be held under the soil

Addition of manure/lime

- Addition of manure helps to build a good soil structure which holds more water

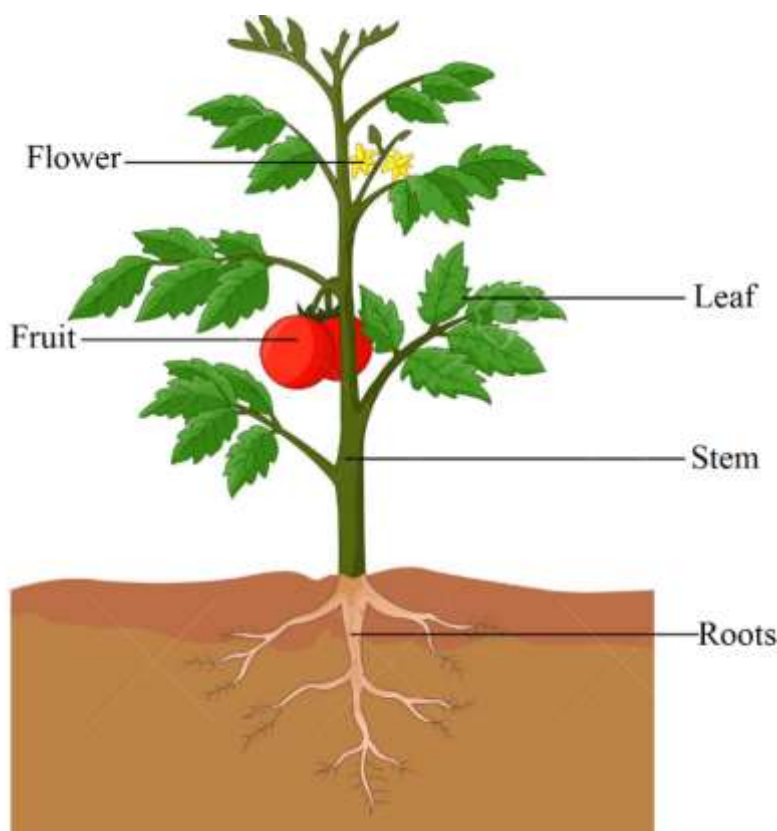
CROP HUSBANDRY

Structure of flowering plants

- Flowering plants produce flowers and seeds then die e.g. sunflower, beans, maize etc.

External parts of flowering plants

- Flowering plants are made up of four basic parts; the roots, stem, leaves and flowers



Function of parts of flowering plants

Part	Functions
Roots	- Absorb water and mineral salts from the soil - Support or anchor the plant firmly into the ground
Stem	- Passage of water and mineral salts from the roots to the leaves and glucose to the rest of the plant - Supports branches, leaves and flowers on the plant
Leaves	- Trap sunlight energy used in <i>photosynthesis</i> - Green leaves manufacture plant food during photosynthesis - Release excess water from the plant through a process called <i>transpiration</i>
Flowers	- Sexual reproductive organs of flowering plants - Produce male (pollen) and female (ovules) sex cells

Classification of plants

- Plant classification refers to the grouping of plants according to their similarities

Classification of plants according to edible parts

Seed crops

- These are crops grown for their edible grain or seeds
- There two common types of seed crops:

Cereals

- Grain crops from grass crops that are grown for their carbohydrate (starch) rich seeds e.g. maize, wheat, rice, sorghum, millet and rapoko

Legumes

- These are grain crops that have pods and produce seeds rich in fats and proteins e.g. groundnuts, roundnuts, beans, peas and acacia

Root crops

- Crops grown for their edible underground roots or rhizomes e.g. carrots, sweet potatoes and beetroot

Tuber crops

- Crops grown for their thickened underground stem e.g. Irish potato and cocoa yams

Leaf crops

- These are crops grown for their edible leaves or stem or flowers e.g. e.g. covo, lettuce, cabbage, celery, broccoli etc.

Fruit crops

- Crops grown for their edible fleshy fruits e.g. tomatoes, eggplant, oranges, mango, peaches etc.

Bulb crops

- These are crops grown for their edible underground swollen leaves e.g. king onions, leak and garlic

Classification of plants according to life cycle

Annual crops

- Crops that complete their life cycle i.e. from seed germination to reproduction in one season e.g. peas, maize, potatoes, pumpkins, rice, lettuce, peas etc.

Biennial crops

- Crops that complete their life cycle in two growing seasons e.g. cabbages, cassava, carrots, onions etc.

Perennial crops

- These are crops that do not die after producing flowers.
- They continue growing from season to season or year to year i.e. they live for many years e.g. cotton, mango, orange and sugarcane

Horticulture

- It is a branch of agriculture which is concerned with garden crops
- Horticulture can be defined as the branch of agriculture concerned with intensively cultivated plants directly used by man for food, medicinal or aesthetic purposes.

Branches of Horticulture

Pomology

- the production of edible fruits e.g. orange, mango, peach, apple etc.

Olericulture

- the production of vegetables e.g. cabbage, peas, carrots, tomatoes, potatoes etc.

Floriculture

- involves the growing, marketing of flowers and other ornamental plants

Decorative horticulture)

- the design and alternation of a portion of land by use of planting material (flowers and other ornamental plants)

Importance of Horticulture

Social importance

- Fruits and vegetables are sources of vitamins, minerals, carbohydrates, fats and proteins
- Flowers and ornamental plants are used for decoration and recreation
- The parts like stem, leaf, flowers, roots and even the fruits of horticulture plants are used to make drugs, chemicals, insecticides, germicides etc. e.g. rose water is used to cure eyes

Economic importance

- Horticultural products are raw materials for processing and canning industries
- Horticulture creates employment through field operations like fruit picking/harvesting, grading, packing, selling etc. and chain of industrial processes
- Exportation of products such as fruits and flowers earn foreign currency to the country
- Many horticultural products are sources of dietary nutrients for humans

Land preparation

- Land preparation is the conditioning (clearing and tillage) of the land before it can be used for crops production.

Reasons for land preparation

- Land preparation is done to:
 - make the land suitable for growing crops.
 - destroy growing plants and weeds.
 - improve soil aeration and water infiltration.
 - improve seed germination.
 - break soil pans and surface capping.
 - level the land.
 - avoid damage of equipment.
 - prepare seed beds.
 - practice irrigation easily.
 - incorporate manure and lime.
 - expose pests to the sun so that they die (to control pest and diseases).

Seedbed preparation

- A seedbed is a small bed in which seedlings are raised
- They are also called nursery beds

Steps in seedbed preparation

Marking

- Measuring the dimensions of the seedbed and pegging the corners
- Seedbeds are usually made 1 to 1.5 metres wide

Digging

- The soil is loosened and turned over using digging tools

Making ridges

- A garden line is used to show where the ridges will be
- A hoe or rake is used to make the ridge under the line

Manuring

- Compost or farmyard manure can be sprinkled over using hands or spade to supply soil nutrients

Levelling

- This is done using a rake to create an even seedbed
- It also mixes manure with soil and breaks soil lumps to create a fine tilth

Crop protection

Pests

- A pest is an organism that is harmful in some way to crops.
- If unprotected, crops will not grow well and yield will be reduced by pests.

Effects of pests to crops

- Some pests are vectors (transmitters) of plant diseases e.g. leaf hopper and aphids
- Pests damage plant tissues through biting, boring or sap sucking
- They also destroy agricultural stored products such as grains
- Pest infestation in fields can result in total crop failure and yield loss
- Pests affect yield quality thus reduce the nutritional and economic value of the crops

Classification of pests

- Pests attack crops in a number of ways and they can be classified according to the type of damage they cause (feeding habits).

Biting and chewing pests

- These are adult insects or their larvae which have biting and chewing jaws called *mandibles*.
- Mandibles cut or bit plant leaves, leaf sheaths and in some cases the young stem which is soft.
- Included in this group are locusts, termites, leaf miners, beetles, leaf miners, armyworms and larvae of moths.
- Damage is by eating leaf material or cutting young stems but there is no transmission of diseases.

Piercing and sucking pests

- These pierce plants with a needle-like mouth parts called *stylets*.
- Examples are aphids, mealy buds, scale insects and red spider mites.
- Piercing does very little damage to plant tissues, the actual damage is through sucking of plant sap.
- Saliva injected into plant may be toxic or contain disease causing micro-organisms especially viruses.
- Aphids are the most notorious, they transmit rosette viral disease in ground nut, leaf roll in potatoes in potatoes and tobacco mosaic virus in tobacco.

Boring pests

- These make holes in the plant stem, fruit or grain and live inside them for some part of their life cycle.
- This group includes grain weevils (adults and larva), maize stalk borer and American bollworm of cotton (*Heliothis spp*).

Nematodes (Eelworms)

- These are very small (microscopic), non-segmented worms, some of which live on root of certain preferred crop plants such as tobacco, potatoes and tomatoes.
- Usually the larva lives in roots which then form swellings called *galls* in response to the attack.

Plant diseases

- A plant disease is an impairment of health (abnormal condition) in plants that is detrimental.

Classification of plant diseases

- Plant diseases are classified according to their causative agents
- They are caused by fungi, bacteria, viruses or nutrient deficiencies

Fungal diseases

- These are diseases caused by fungi.
- Fungi are simple plants without chlorophyll which range from microscopic to large mushrooms.
- Disease causing fungi cannot photosynthesise and obtain their food from plants
- Examples include late blight of potatoes, damping-off of seedlings, powdery mildew of beans and peas and downy mildew of peas
- Fungi spread by means of spores which can germinate and infect plants through the stomata or wounds

Bacterial diseases

- These are caused by bacteria.
- Bacteria are microscopic single celled organisms and are smaller than fungi.
- Bacteria cause fewer diseases as compared to fungi.
- The spread of bacterial diseases is either by rain splash, insects, or man and machinery.
- Examples of bacterial diseases are bacterial blight of cotton (Black arm or Angular leaf spot) and bacterial wilt of tomatoes

Viral diseases

- These are caused by viruses.
- Viruses are very minute pathogens which cannot be seen through a common microscope but through an *electron microscope*.
- They affect plants by destroying chlorophyll resulting in yellowing of the plant and causes stunting (reduces growth) of crops.
- Viral diseases in plants are spread by vectors such as aphids, leaf hoppers, thrips and white flies or by planting infected material during vegetative propagation or by tools and hands of workers
- Examples include maize streak virus disease of maize, tobacco/tomato mosaic virus, bushy top and rosette virus of groundnuts

Weeds

- Weeds are undesirable or unwanted plants which grow in crop fields
- They are competitive and adaptable to adverse conditions or environments

Harmful effects of weeds

Harmful effects of weeds are divided into direct and indirect effects.

Direct effects

- They reduce crop yields through competition for light, moisture and nutrients.
- Weeds interference with agricultural operations such as cultivation, spraying and harvesting
- Some weeds such as *Shamva grass* have hairs on their surfaces which cause itching when they come in contact with the skin and this disturbs workers.
- Weeds can also reduce machine efficiency through blockages.
- Some weeds can reduce quality of harvest e.g. black jack in cotton and too much weed seed in crop harvest used for seed leads to rejection of the seed.
- Weed burs of weeds like upright star bur, black jack and pig weed reduces quality of cotton.
- Weeds increase the cost of production because money has to be spent to control the weeds either chemically or mechanically.
- Some weeds are poisonous to livestock e.g. *Lantana camara*.
- Aquatic weeds reduce affects the quality of water for irrigation in rivers and dams e.g. the water hyacinth in Manyame river at lake Chivero.

Indirect effects

- Weeds harbour rodents e.g. rats and mice which destroy crops.
- They also host plant pests and diseases e.g. Wild rose (*Hibiscus dongolensis*) hosts the cotton boll worm

Benefits of weeds

- They provide cover against soil erosion during furrow periods in a crop rotation cycle.
- Weeds add organic matter to the soil upon decomposition.
- Some weeds are used as human food e.g. Amaranthus (mowa/bonongwe), black jack (tsine) and okra.
- Other weeds have medicinal properties and are used to cure snake bites, gastric troubles, skin disorders and some control diabetes
- Provide nectar for bees.

Classification of weeds

Annual weeds

- These are weeds which complete their life cycle (grown, mature and produce seeds) in one season e.g. Black jack, Mexican marigold, Sodom apple and Witch weed.

Perennial weeds

- These need more than one season to complete their life cycle e.g. Wondering jew, Coach grass, Star grass, Thorn apple and Nutgrass.

ANIMAL HUSBANDRY

Types of livestock

- Livestock refers to the various kinds of domesticated animals reared on a farm.

Poultry

- Includes all animals in the bird family
- These animals are covered by feathers, lay eggs and are two legged
- Poultry includes animals such as chickens, ducks, turkeys, geese, guinea fowl and pigeons

Importance of poultry

- Most birds are kept for meat e.g. broilers, turkeys, ducks
- Some are kept for eggs e.g. layers, guinea fowl
- Poultry are a source of income when sold as live birds or dressed chicken
- Poultry production creates employment

Fish

- These are aquatic (water living) animals
- Fish are grown in tanks, ponds, cages or irrigation ditches
- There are different types of fish breeds farmers can produce, these include breams, tilapia and trout

Importance of fish

- Source of proteins, vitamins and minerals for human beings
- Also provide proteins to animals when fed to animals as fish meal feed formulation
- Good source of income when sold as fresh or dried
- Fish are raw materials for processing and canning industries

Ruminants and non-ruminant animals

Ruminants

- These are animals which graze grass and browse on tree leaves
- They have a stomach chamber which is divided into four compartments: rumen, reticulum, omasum and abomasum
- They chew cud after grazing, during grazing or browsing, the grass or leaves collect in the first stomach then is later returned to the mouth for chewing, a process called *regurgitation*
- Ruminants are able to digest *cellulose*, a tough plant material
- These animals include cattle, sheep and goats

Non-ruminants

- These are animals which have one stomach chamber and do not chew the cud
- Non-ruminants cannot digest cellulose and thus feed on foods with less fibre
- These animals include rabbits, donkeys, pigs and horses

Importance of ruminants and non-ruminants

- Produce meat e.g. cattle, sheep, goats, rabbits and pigs
- Produce milk e.g. dairy cattle and goats
- Provide raw materials for processing industries for the production of processed foods such as cheese, ice cream, yoghurt, and other products such as shoes from the hide
- Source of income when the animals or meat is sold
- Creates employment through the production system

Animal health

- Good health is a condition when an animal is in a good state and responding well to the environment
- *Ill*-health is a condition in which the animal is not well
- Healthy and *Ill*-health animals can be identified by their behaviour and condition such as physical appearance

Healthy animal	<i>Ill</i> -health animal
Has normal posture, very alert and bright eyes	Keeps down with running nose and dull eyes
Walks properly with regular steps	Difficulty walking with irregular steps and usually unbalanced
Has a good appetite and feed to satisfaction	Loss of appetite and does not feed
Normal colour of urine, dung or droppings	Stained or abnormal colour in urine, dung or droppings
Normal breathing	Abnormal breathing and continuous coughing
Correct consistency of dung or droppings not too hard or watery	Hard or watery dung/droppings

Small livestock production

Broiler production

Importance of rearing broilers

- Broilers produce meat which is an important source of high-quality proteins, minerals and vitamins
- Broilers have a high feed conversion ratio, they grow quickly per mass of feed consumed
- Broilers are an important source of income, either sold as live or dressed birds. When exported, broilers give farmers foreign currency
- Broiler production creates employment throughout the production system
- Broiler droppings have high nutrient content and are used as manure in crop production
- Feathers of broilers can be used as dusters or used to make pillows

Breeds of broilers

- There are different breeds of broilers
- Common broiler breeds in Zimbabwe are:
 - Hubbard
 - Cob 500
 - Ross

Housing requirements for broilers

- Broiler chicks are received from hatcheries when they are a *day old*
- Common hatcheries in Zimbabwe are Irvines, Pro feeds and Novatek
- After arrival the chicks should be kept in *brooders*
- Brooders are special housing for raising chicks which provide adequate warmth needed by chicks until they are two or three weeks old
- The common types of brooders are *canopy brooder*, *hay box brooder* and *circular wire mesh brooder*
- Brooders also protect the chicks from infection, diseases and parasites
- Brooders should be kept clean and well ventilated to prevent diseases and parasites

Brooder temperature management

- Warmth in a brooder is provided through the use of infra-red lamps, tungsten bulbs, charcoal burners, electric heaters or gas heaters
- Poor temperature management can result in high deaths (mortality) during the brooding period
- Litter material such as soft grass, wood shavings or sawdust should be spread to a depth of 5cm on the floor of the brooder to keep the place warm
- The chicks require a warm temperature of 35°C which needs to be reduced by 2°C every week as they grow
- When heat is inadequate, chicks will huddle together, this indicates that the chicks are cold
- If using infra-red lamp, the farmer should add an extra infra-red lamp or lower the height the source of heat or move it closer to the chicks

- When temperatures get too high in the brooder, the chicks respond by moving away from the source of heat
- The farmer should increase the distance between source of heat and the chicks

Deep litter housing for broilers

- After the brooding period chicks are introduced to a deep litter housing system
- Deep litter housing systems involve rearing birds in large unit with no divisions or cages and bedding made from wood shavings, saw dust or dry grass is laid on the floor
- The birds live and feed from drinkers (drinking water containers) and feeders (feed containers)
- The feed and water containers are changed to larger sized feeding equipment to accommodate the age of the broilers
- Clean water should be available all the time
- Spilling of water should be avoided as it promotes the multiplication of pathogens
- The bedding should be changed regularly to keep it dry and prevent diseases

FARM STRUCTURES AND MACHINERY

Farm implements

Tillage implements

- Tillage is the mechanical preparation of the soil e.g. ploughing, digging, disking, ripping, cultivation etc.
- Tillage operations are divided into two phases:
 - Primary tillage
 - Secondary tillage

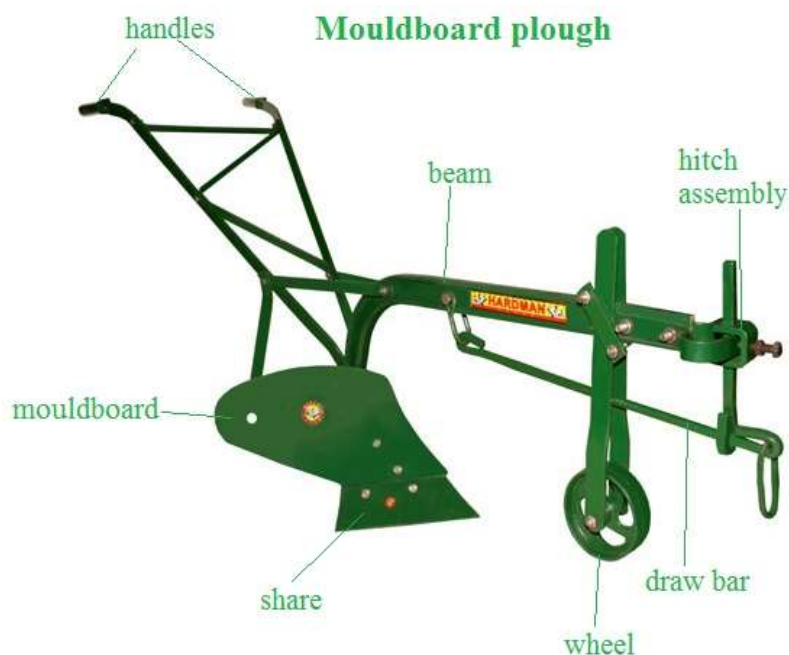
Primary tillage

- This is the initial cultivation of the soil and involves the use of heavy-duty equipment.
- It leaves the soil in a rough state not ideal for planting.
- Primary tillage implements are:
 - Disc plough
 - Mouldboard plough
 - Chisel plough
 - Ripper

Secondary tillage

- This is a light duty operation specifically to break large clods of soil produced by primary tillage.
- It leaves the soil level, with a fine tilth suitable for planting.
- Secondary tillage implements are:
 - Disc harrow
 - Cultivator
 - Spike-toothed harrow
 - Planter
 - Roller

Mouldboard plough



Functions of parts of a mouldboard plough

Part	Function
Share	Penetrates and cuts the soil in a horizontal direction below the soil surface leaving a furrow
Mouldboard	Receives furrow slices from the share and turns them over
Landside	Slides along the furrow wall to stabilize the plough while in operation
Frog	It is part of the plough bottom on which the share, mouldboard and landside are attached rigidly
Hitch assembly	It is used to adjust depth and width of furrow It is where the trek chain is attached
Draw bar	Transmits the pull of oxen to the beam Can be raised or lowered at the front to control depth of ploughing
Wheel	Makes the plough steady and stable Also prevents the plough from digging too deep
Handles	These are used to control and guide the plough
Beam	It is where the frog, handles and hitch assembly are attached

Fencing

- A fence is a barrier constructed around property, fields, paddocks or animal kraals

Reasons for fencing

- To protect crop fields from livestock
- To make paddocks in order to facilitate rotational grazing, proper breeding systems and night paddocking.
- To control movement of animals and people around the farm.
- To mark farm boundaries in order to avoid boundary disputes.
- To keep different classes of animals separate e.g. bulls alone, cows, heifers, steers etc. this makes feeding of the animals easier.
- To enable mixed farming at the farm.
- To protect property from thieves.

Types of fences

1. Barbed wire fence
2. Plain wire fence
3. Post and rail fence
4. Diamond wire mesh fence
5. Hedge fence
6. Stone wall fence
7. Electric fence

Harnessing

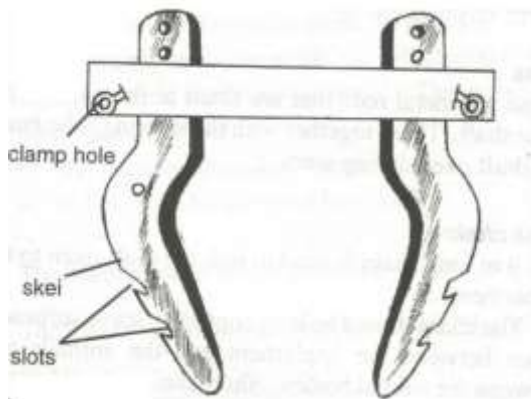
Harnesses

- Harnesses can be defined as the straps, bands and wooden devices that link draft animals and implements.
- They also serve to control and direct draught animals during operation of implements
- Harnesses for horses, donkeys and mules are very different from harnesses used on cattle
- Cattle produce most of their power from their necks and humps that is why farmers fit yokes around necks of trek oxen.

Types of yokes

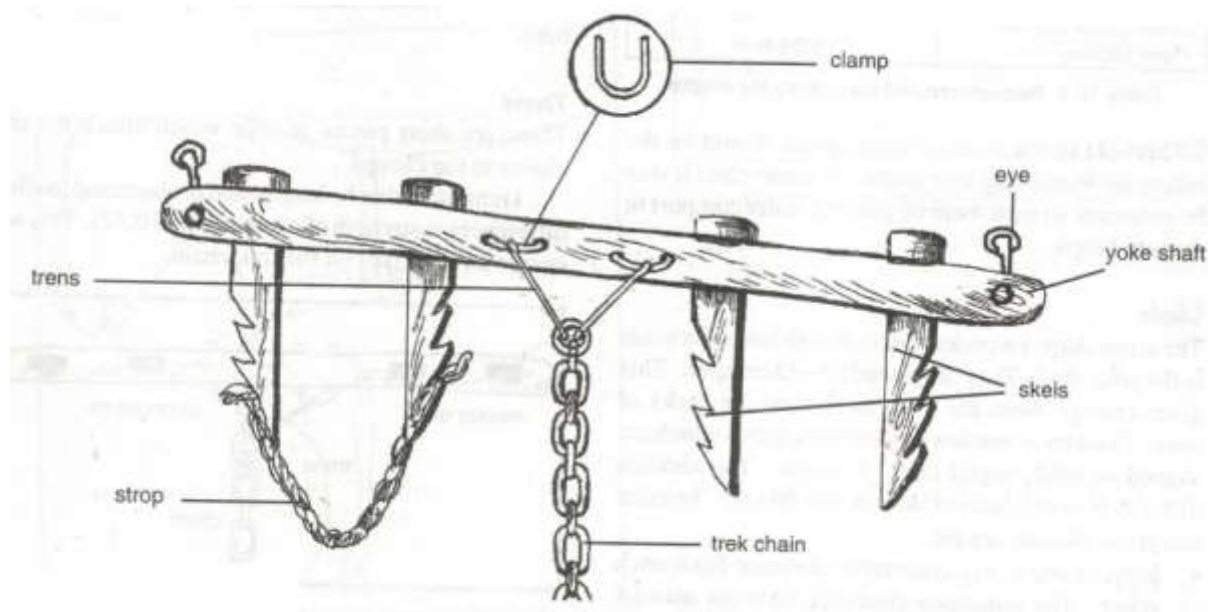
1. Single neck yoke

- This is a wooden framework used to secure a single draught animal to farm implements



2. Double neck yoke

- This is a wooden framework used to link two draught animals and secure them to farm implements



Functions of parts of a double neck yoke

Part	Description and functions
Yoke shaft	This is the main framework of the yoke that carries or bears load Its length is determined by use of the yoke
Skeis	These are wooden pieces shaped to fit into the yoke shaft slots and have notches on the outer edges which hold the looped ends of strops They keep animals secured at a distance from each other Also prevent the trek chain from continuously rubbing on to the skin of animals
Strops	These are made from hides, robes and a combination of rubber and fibre They have looped ends that fit into the notches on skeis Strops prevent the yoke from slipping back over draught animals
Clamps	These are U-shaped metal rods fitted at the centre of the yoke They are the point of attachment for trens
Trens	These are large metal rings that link the trek chain to the clamps
Trek chain	It is made of several metal rings joined together to a length of 2.9 metres The trek chain is fitted to the clamp and links draught animals to the implement

Types of double neck yokes

Type of yoke	Distance between animals	Drawn farm implements
Plough yoke	90 cm	Plough, ripper and planter
Cultivator yoke	180 cm	Cultivator and ridger
Cart yoke	120 cm	Cart

AGRI-BUSINESS

Farm records

- Farm records are documents that show detailed and accurate information (facts and figures) of everything connected with the farm.

Importance of farm records

- They show history of the farm
- Farm records are used to control and track movement of tools and equipment into and out of the farm
- They help to identify losses of stock or equipment
- Also help in planning and implementing irrigation programmes
- They are used as source documents for the calculation of profit or loss made on the farm
- Farm records are required by banks when securing loans or funds from donors
- A farmer uses farm records as a reference in decision making
- Help in evaluating the performance of individual stock
- They are used in breeding of animals to keep track of mating dates, age and breeds used
- Records are useful guidelines for preparing budgets

Types of farm records

Physical records (Production/Management records)

- These show farm activities and contain all the details in the production of crops or livestock
- Livestock records include breeds, mating dates, birth dates, weaning dates, weaning mass, breeding season, dipping, mating ratios, supplementary feeds, culling
- Crop records include ploughing dates, irrigation dates, sowing dates, fertiliser application dates, seeds, seed rates, cultivators, harvest time, chemicals used for pest and disease control, land area, marketing
- Physical records also include inventory records (records of assets or tools) and weather records (rainfall and temperature)

Financial records

- These records show or have monetary values of farm activities, inputs and outputs
- They are records of the income (revenue) and expenditure on a farm
- Financial records include:
 - Income and expenditure account
 - Profit and loss account

Agricultural co-operatives

- A co-operative is a registered organization of people who have decided to work together for mutual benefits.
- The minimum number of members for a co-operative is 10.

Principles of co-operatives

- Anyone who meets the requirements of the co-operative is free to join (open membership)
- In a co-operative, there is democratic control over assets, operations and decision-making processes
- Profits are distributed to members following ratios agreed to by all members usually based on each member's contribution
- All members are obliged to have full knowledge of the functioning and benefit offered by the co-operative

Types of agricultural co-operatives

Producer co-operative

- Farmers grow crops or produce livestock together
- Member in this co-operative must contribute equal amounts of efforts in work and time

Marketing co-operative

- Producers sell their products together to receive better prices for their products and meet supply needed by a market

Savings co-operative

- Interested farmers can group together and establish an account where members deposit an agreed amount of money for a certain period of time
- Members can decide on sharing the money or they can use it to buy inputs and share the inputs